On the Asymptotic Density of Convergent Residue Classes in the Collatz Map

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Abstract

We study the dynamics of the semi-accelerated Collatz map

$$g(x) = \begin{cases} \dfrac{x}{2} & \text{if } x \text{ is even,} \\ \dfrac{3x+1}{2} & \text{if } x \text{ is odd.} \end{cases}$$

For structural analysis we relate g to the fully accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

Here $v_2(n)$ denotes the 2-adic valuation of n (largest k such that $2^k \mid n$).

Our approach decomposes the positive integers into residue classes determined by finite parity patterns under the fully accelerated map f, which serves as a symbolic compression of trajectories of the semi-accelerated map g. Trajectories are encoded as parity words (sequences of division counts $v_2(3x+1)$), yielding associated affine maps that describe the finite-step action of f on entire classes at once. This framework allows systematic enumeration of convergent residue classes and quantitative control of their combinatorial growth via a grammar derived from lower bounds on the total number of divisions by two required for admissibility.

Comparing this growth with the powers of two governing residue-class densities, we prove that the cumulative natural density of integers belonging to residue classes with finite stopping time approaches one. Consequently, almost every positive integer eventually reaches a smaller value under iteration of the map (in the sense of natural density). While zero-density exceptional trajectories are not excluded, the results provide a measure-theoretic description of typical Collatz behavior.

Contents

1 Introduction

The $3x+1$ problem, also known as the Collatz conjecture [1], concerns the behavior of the map

$$T(x) = \begin{cases} \dfrac{x}{2} & \text{if } x \text{ is even,} \\ 3x+1, & \text{if } x \text{ is odd.} \end{cases}$$

on the positive integers. It asserts that every positive integer eventually reaches 1 under iteration of T. Despite its elementary formulation, the conjecture has resisted proof for decades [2, 3]. Extensive computational verification confirms convergence for all integers up to very large bounds [4], but such verification does not constitute a proof and offers limited insight into the global structure of Collatz trajectories.

A useful intermediate form is the *semi-accelerated map*

$$g(x) = \begin{cases} \dfrac{x}{2} & \text{if } x \text{ is even,} \\ \dfrac{3x+1}{2} & \text{if } x \text{ is odd.} \end{cases}$$

which applies a single division by two immediately after each odd step. This map preserves the essential dynamics while simplifying the combinatorial structure of trajectories.

For structural analysis it is convenient to work with the *fully accelerated map*

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

defined on odd integers. This map removes all powers of two in a single iterate and satisfies

$$f(x) = g^{\,v_2(3x+1)}(x),$$

where the first iterate applies the odd branch and the remaining iterates are divisions by two. All symbolic encodings and residue-class constructions in this paper are formulated in terms of the fully accelerated map.

Our approach is to study the *structural properties* of Collatz trajectories rather than individual integers. Specifically, we partition the positive integers into *residue classes* determined by finite sequences of division counts following odd steps under the fully accelerated map f—called *parity words*. Each admissible parity word corresponds to a finite segment of a trajectory under f, terminating at the first step where the iterate is strictly smaller than its starting value.

We then associate an *affine map* to each parity word, which compresses the trajectory segment into a single linear-affine operation modulo a power of two (determined by the total number of divisions). This representation captures the multiplicative and additive effects of the trajectory while abstracting away the starting integer, allowing reasoning at the level of classes rather than individual numbers.

To control growth, we introduce an *increment grammar* derived from the minimal total number of divisions by two required for admissible parity words. This grammar encodes allowable patterns of growth in these denominators. It will later underpin a canonical *horizontal-sum construction*, which enumerates convergent classes and controls their combinatorial growth.

This perspective allows us to show that almost every positive integer eventually reaches an iterate strictly smaller than its initial value under iteration of the fully accelerated map f, in the sense of natural density. While this framework does not exclude the logical possibility of zero-density exceptional trajectories, it provides a strong quantitative understanding of typical Collatz dynamics and highlights the combinatorial and arithmetic mechanisms that govern convergence for the vast majority of integers.

Structure of the Paper. Section 2 introduces stopping time, convergent segments, and residue classes. Section 3 defines parity words, total division counts, and their associated densities. Section 4 develops the parity-word grammar and admissibility conditions, leading to affine representations of residue classes. Section 5 presents the increment grammar and horizontal-sum construction, which enumerates convergent classes and controls their combinatorial growth. Section 7 applies this machinery to establish vanishing density, while Section 8 completes the argument by proving global descent.

2 Definitions and Preliminaries

2.1 Stopping Time

Definition 2.1. The *stopping time* $\sigma(x)$ of a positive integer x (sometimes called the finite stopping time or local descent time) is the minimal integer $k \geq 1$ (if such a k exists) such that

$$f^k(x) < x.$$

If no such k exists, we write $\sigma(x) = \infty$. Stopping time behavior has been studied extensively in the context of the $3x + 1$ problem [3].

2.2 Convergent Segments

Definition 2.2. A *convergent segment of length* k is a finite sequence

$$x, f(x), f^2(x), \ldots, f^k(x)$$

such that $f^k(x) < x$ and k is minimal. No assumption is made about eventual convergence to 1; only strict descent at step k, i.e., $k = \sigma(x)$, is required. Finite segments of this type were considered in the context of the accelerated Collatz map by Everett [5].

2.3 Residue Classes

Encoding trajectories via residue classes or arithmetic progressions has appeared in prior structural approaches to the $3x + 1$ problem [6, 7].

Iterates are indexed starting from 0, so that $f^0(x) = x$. Integers x and y are equivalent modulo depth k if the sequence of parities of $f^i(x)$ and $f^i(y)$ agree for $i = 0, \ldots, k-1$. In other words, they share the same parity sequence. This convention aligns with the 0-based combinatorial and OEIS sequences used in the analysis.

Definition 2.3 (Residue Class of Depth k)**.** Fix $k \in \mathbb{N}$ and $x \in \mathbb{N}$. Let $\mathrm{par}(n) = n \bmod 2$. The *residue class of depth* k *with respect to* f containing x is

$$[x]_k := \{y \in \mathbb{N} \mid \mathrm{par}(f^i(y)) = \mathrm{par}(f^i(x)) \text{ for } i = 0, 1, \ldots, k-1\}.$$

Here $[x]_k$ encodes the parity sequence; the associated parity word ω further records the number of divisions by two following each iterates of f, as formalized in Section 3.

Remark (Arithmetic structure of residue classes)**.** Later (Section 4), we show that for admissible parity words, the residue class associated to a parity sequence of depth k is either empty or an arithmetic progression modulo a power of two determined by the total division count $D(\omega)$. At this stage, it suffices to know that $[x]_k$ encodes a parity sequence; any further arithmetic structure will be derived in subsequent sections.

Definition 2.4 (Convergent Residue Class)**.** A residue class $[x]_k$ is called *convergent of stopping time* k if

1. $f^k(y) < y$ for all $y \in [x]_k$, and
2. no smaller integer $j < k$ satisfies $f^j(y) < y$ for any $y \in [x]_k$.

Here k is minimal with this property for elements in $[x]_k$. Uniformity across the class follows from the fact that all elements share the same parity sequence; this will be formalized and proved in Section 4.

Remark. This definition is stronger than requiring $\sigma(x) = k$ for a single representative x, since it enforces uniform descent at step k for all elements of the residue class.

3 Division Counts and Path Encoding

The structure of convergent residue classes is governed by the total number of divisions by two incurred along an accelerated Collatz trajectory. At each step of the accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}},$$

all powers of two are removed from $3x+1$. The number of removed factors of two at the i-th iterate is therefore the 2-adic valuation

$$d_i := v_2(3f^i(x)+1).$$

Here d_i corresponds to the division count in the transition $f^i(x) \to f^{i+1}(x)$. Since the fully accelerated map always produces odd iterates, $3f^i(x)+1$ is even and hence $d_i \geq 1$.

3.1 Formal Definition of Parity Words

In this section we introduce a symbolic encoding of Collatz trajectories based on the number of divisions by two removed at each step of the accelerated map. We call such an encoding a *parity word*. The precise combinatorial constraints determining which parity words correspond to actual trajectories are deferred to Section 4; however, the definitions given here are sufficient for the analysis of residue class densities.

Let a Collatz path be a sequence of iterates under the accelerated map

$$f(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

Definition 3.1 (Parity Word)**.** A *parity word* of length $m \geq 0$ is a finite sequence

$$\omega = (d_0, d_1, \ldots, d_{m-1}),$$

where each d_j is a positive integer representing the number of divisions by two applied at the j-th iterate of the accelerated map. Admissibility constraints on which sequences can occur are defined in Section 4.

For $m = 0$, the word is empty and represents the *trivial zero-step segment.*

Lemma 1 (Uniform Valuation on Residue Classes)**.** Let a residue class be determined by a parity word of length $m > 0$ with division counts $(d_0, \ldots, d_{m-1})$. Then for every integer in the class, the division counts d_i are identical for all $0 \leq i < m$.

Remark. The parity word determining the residue class is understood to terminate at length m. No claim is made about parity symbols or division counts beyond this prefix, which lie outside the equivalence relation defining the class.

3.2 Affine Maps Associated to Parity Words

Each parity word formally determines an affine map of any integer x:

$$x \longmapsto \frac{3^m x + c(\omega)}{2^{d_0+\cdots+d_{m-1}}}, \tag{3.1}$$

obtained by symbolic composition of the accelerated map along the word. Here $c(\omega)$ is a canonical integer uniquely determined by the parity word (see Appendix A). This representation encodes the arithmetic effect of the sequence of accelerated steps encoded by the word on any starting integer.

Remark (Parity Words Encode Structure, Not Values)**.** Fixing a parity word ω determines the sequence of arithmetic operations applied along an accelerated Collatz trajectory. Consequently, both the linear coefficient $3^m/2^{d_0+\cdots+d_{m-1}}$ and the affine term $c(\omega)$ depend only on ω and not on the starting integer.

This allows reasoning about convergence or divergence at the level of residue classes without reference to specific integers.

Remark (Residue Classes and Parity Words)**.** Each admissible parity word ω determines a unique arithmetic progression modulo $2^{d_0+\cdots+d_{m-1}}$.

3.3 Total Division Count

Let a parity word $\omega = (d_0, \ldots, d_{m-1})$ encode a segment consisting of m iterates of the accelerated map f. Each iterate of the fully accelerated map removes a finite number of factors of two from $3x+1$. This number is precisely the entry of the parity word corresponding to that iterate.

Definition 3.2 (Total Division Count)**.** For a parity word $\omega = (d_0, \ldots, d_{m-1})$ encoding the m iterates of f of a path, define its *total division count* by

$$D(\omega) := \sum_{j=0}^{m-1} d_j,$$

where d_j equals the 2-adic valuation occurring at the j-th iterate for any trajectory realizing ω.

Thus $D(\omega)$ is precisely the exponent of 2 appearing in the denominator of the linear-affine form of the m-th iterate of f (see Equation (3.1)).

Parity words with the same length m may differ in the distribution of divisions by two between steps and hence in their total division count. However, for each parity word, its total division count $D(\omega)$ is uniquely determined by the sequence of divisions encoded in the word.

Definition 3.3 (Minimal Total Division Count)**.** Let $A020914(m)$ denote the term of the OEIS sequence A020914, which is the minimal total number of divisions by two among all *admissible* parity words containing exactly m iterates of f.

This quantity is given by $\lceil m \log_2(3) \rceil$, a fact derived in Section 4 and tabulated in A020914. Equivalently, it is the smallest integer D such that $2^D \geq 3^m$. Intuitively, each odd step multiplies the trajectory by 3, so the cumulative divisions by two must at least balance this growth; hence the minimal total division count corresponds to the smallest exponent of 2 that offsets the factor of 3^m in the affine map for a path with m iterates of f.

3.4 Density of Residue Classes

Lemma 2 (Disjointness of Residue Classes)**.** Fix $m \geq 0$. The residue classes induced by distinct parity words $\omega = (d_0, \ldots, d_{m-1})$ with m iterates of f are pairwise disjoint.

Proof. Let ω be a parity word with m iterates of f and total division count

$$D(\omega) = \sum_{j=0}^{m-1} d_j.$$

It determines an affine map

$$x \longmapsto \frac{3^m x + c(\omega)}{2^{D(\omega)}}.$$

For the m-th iterate of f to be an integer, x must satisfy the integrality condition

$$3^m x + c(\omega) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^m is odd, it is coprime to $2^{D(\omega)}$. Hence it is invertible modulo $2^{D(\omega)}$; that is, there exists an integer $(3^m)^{-1} \bmod 2^{D(\omega)}$ such that

$$3^m \cdot (3^m)^{-1} \equiv 1 \pmod{2^{D(\omega)}}.$$

Multiplying the congruence by this inverse yields a unique residue class

$$x \equiv r_\omega \pmod{2^{D(\omega)}}.$$

Suppose two distinct parity words $\omega \neq \omega'$ produced the same residue class. Then their integrality conditions would define the same affine map

$$x \mapsto \frac{3^m x + c}{2^D}.$$

However, the symbolic composition of the accelerated map is unique: at each stage, the exponent d_i is exactly $v_2(3f^i(x) + 1)$, so the sequence of exponents is uniquely determined by the successive 2-adic valuations along the trajectory, and hence by the parity word itself. Therefore $\omega = \omega'$, a contradiction.

Consequently, distinct parity words determine distinct and disjoint residue classes. □

Proposition 3.1. A residue class induced by a parity word ω has natural density

$$2^{-D(\omega)}.$$

In particular, for classes with m iterates of f,

$$\text{density} \leq 2^{-A020914(m)}.$$

Proof. If $m = 0$, then the parity word is empty and $D(\omega) = 0$, so the residue class contains all integers and has natural density $1 = 2^0$.

For $m > 0$, each residue class is an arithmetic progression with modulus $2^{D(\omega)}$ and exactly one representative per period. The natural density of such a class is therefore $2^{-D(\omega)}$. □

Remark. For fixed m, there are finitely many admissible parity words with m iterates of f. Let $\widetilde{A}(m)$ denote the number of such words that correspond to convergent residue classes. Since the classes are pairwise disjoint, their total density contribution at iterate depth m is bounded by

$$\widetilde{A}(m)\, 2^{-A020914(m)}.$$

3.5 Example

Consider convergent segments with $m = 4$ iterates of f. The minimal total division count for $m = 4$ is

$$A020914(4) = 7.$$

One admissible parity word attaining this minimum is

$$\omega = (1, 1, 1, 4),$$

for which

$$D(\omega) = 1 + 1 + 1 + 4 = 7.$$

Composing the accelerated map symbolically yields

$$f_\omega(x) = \frac{3^4 x + c(\omega)}{2^7} = \frac{81x + 65}{128},$$

so the canonical affine constant for this word is

$$c(\omega) = 65.$$

The integrality condition

$$81x + 65 \equiv 0 \pmod{128}$$

determines a unique residue class

$$x \equiv 15 \pmod{128}.$$

Hence all integers in this class have natural density 2^{-7}.

Interpretation. The word $\omega = (1, 1, 1, 4)$ means:

- At each of the first three iterates of f, $3x + 1$ contains exactly one factor of 2.
- At the fourth iterate of f, $3x + 1$ contains four factors of 2.

Under the accelerated map

$$f(x) = \frac{3x + 1}{2^{v_2(3x+1)}},$$

all of these divisions occur within a single iterate. Thus $d_3 = 4$ means that the fourth iterate divides by 2^4 in that step. The constant $c(\omega)$ is fixed entirely by this symbolic structure and ensures that precisely the integers in the residue class $x \equiv 15 \pmod{128}$ produce integer iterates under the affine form.

This example illustrates that convergence properties depend only on the parity word ω and its total division count $D(\omega)$, not on the specific starting value.

4 Grammar of Collatz Parity Words

4.1 Parity Words as Dynamical Encodings

Remark (Parity Grammar as a Dynamical Encoding)**.** Recall from Section 3 that a *parity word* $\omega = (d_0, \ldots, d_{m-1})$ records the exact 2-adic valuation $d_i = v_2(3x_i + 1)$, occurring at each iterate of the fully accelerated Collatz map.

The parity-word grammar introduced in this section provides a symbolic encoding of accelerated Collatz dynamics, following the symbolic approach of Chamberland [7]. This encoding is not injective: multiple positive integers can correspond to the same parity word. Specifically, distinct integers whose trajectories share the same sequence of division counts after m iterates of the accelerated map f trace the same parity word, forming the arithmetic residue classes described in Section 3.

Unlike a purely combinatorial encoding, parity words retain essential dynamical information. Each parity word $\omega = (d_0, \ldots, d_{m-1})$ determines the sequence of multiplicative factors of the linear coefficient

$$r_i = \frac{3}{2^{d_i}},$$

so that the contractive or expansive contribution of each iterate is preserved in the linear coefficient (with the affine term handled separately via $c(\omega)$). This structure will allow us to quantify densities of residue classes and to formulate sufficient conditions for descent, even though individual trajectories are collapsed under the encoding.

4.2 Admissibility of Parity Words

Definition 4.1 (Admissible Parity Word)**.** Let ω be a parity word with m iterates of f and total division count $D(\omega)$. Associated to ω is the affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

where $c(\omega)$ is the canonical integer determined by ω.

The parity word ω is said to be *admissible* if there exists at least one integer x such that $F_\omega(x)$ is an integer. Equivalently, ω is admissible if the numerator $3^m x + c(\omega)$ is divisible by $2^{D(\omega)}$ for at least one residue class of x mod $2^{D(\omega)}$.

Remark. Admissibility is therefore an arithmetic condition on the word ω itself, expressed as a divisibility constraint on the associated affine map. When this condition holds, the set of all integers x for which $F_\omega(x)$ is integral forms a unique residue class modulo $2^{D(\omega)}$, corresponding to the residue class induced by ω.

Lemma 3 (Minimal Division Count)**.** For any admissible parity word $\omega = (d_0, \ldots, d_{m-1})$ of length m,

$$d_0 + \cdots + d_{m-1} \ \geq \ A020914(m),$$

where

$$A020914(m) = \lceil m \log_2(3) \rceil.$$

Equality holds if and only if ω achieves the minimal possible total division count among admissible words of length m.

Definition 4.2 (Convergent Parity Word)**.** An admissible parity word $\omega = (d_0, \ldots, d_{m-1})$ is said to be *convergent* (in the linear sense) if

$$\frac{3^m}{2^{d_0+\cdots+d_{m-1}}} < 1.$$

Remark (Role of the Affine Term)**.** The constant term $c(\omega)$ in the affine representation

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{d_0+\cdots+d_{m-1}}}$$

encodes the cumulative effect of the additive term inherited from the classical $3x+1$ map along the trajectory determined by ω. Admissibility ensures that this affine map corresponds to genuine Collatz dynamics, mapping a nonempty arithmetic progression of integers to integers. Once admissibility is established, the condition

$$\frac{3^m}{2^{d_0+\cdots+d_{m-1}}} < 1$$

characterizes the asymptotic behavior of the map: the associated linear factor is *contractive*. Thus, although the additive term affects the finite-scale behavior of individual trajectories, the criterion for convergence depends only on the multiplicative component of the affine map.

4.3 Residue Classes Induced by Parity Words

Fix a parity word $\omega = (d_0, \ldots, d_{m-1})$ of length m that is admissible in the sense of Definition 4.1. Let $D(\omega)$ denote its total division count. By admissibility, $D(\omega) \geq A020914(m)$.

Lemma 4 (Residue Class Modulus)**.** Let $x, y \in \mathbb{N}$ both realize the same parity word ω, i.e., their trajectories under f exhibit the division counts encoded by ω for the first m iterates of the accelerated map f. Then

$$x \equiv y \pmod{2^{D(\omega)}}.$$

Proof. For any integer z realizing ω, the m iterates of the accelerated map f encoded by ω induce the affine relation

$$F_\omega(z) = \frac{3^m z + c(\omega)}{2^{D(\omega)}}.$$

By admissibility, $F_\omega(z) \in \mathbb{N}$ is integral. This integrality immediately implies that

$$3^m z + c(\omega) \text{ is divisible by } 2^{D(\omega)}.$$

If both x and y realize ω, then

$$3^m x + c(\omega) \equiv 0 \equiv 3^m y + c(\omega) \pmod{2^{D(\omega)}},$$

which implies

$$3^m (x - y) \equiv 0 \pmod{2^{D(\omega)}}.$$

Since 3^m is invertible modulo $2^{D(\omega)}$, it follows that

$$x \equiv y \pmod{2^{D(\omega)}}.$$

□

Lemma 5 (Associated Affine Collatz Map)**.** Each admissible parity word ω induces a unique affine map

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}},$$

where $c(\omega)$ is uniquely determined modulo $2^{D(\omega)}$ by the requirement that $F_\omega(x) \in \mathbb{N}$ for all integers x in the residue class induced by ω.

Remark (Affine Compression of Trajectories)**.** For any integer realizing a parity word ω, the value $F_\omega(x)$ coincides exactly with the result of applying the accelerated Collatz map through the m iterates of the accelerated map f encoded by ω, including all intermediate divisions by two. Thus F_ω represents a compression of a finite Collatz trajectory into a single affine step.

Lemma 6 (Convergent Residue Class)**.** Each convergent parity word ω induces a unique arithmetic residue class

$$x \equiv r_\omega \pmod{2^{D(\omega)}},$$

whose elements all follow the same parity pattern for m iterates of the accelerated map f and satisfy

$$F_\omega(x) < x \quad \text{for all} \quad x > \frac{c(\omega)}{2^{D(\omega)} - 3^m}.$$

Consequently, every element of the class reaches a strictly smaller integer after completing the m iterates of the accelerated map f encoded by ω.

Remark (Solving the Affine Inequality)**.** The bound $x > \dfrac{c(\omega)}{2^{D(\omega)} - 3^m}$ is obtained by solving the affine inequality $F_\omega(x) < x$ for x:

$$\frac{3^m x + c(\omega)}{2^{D(\omega)}} < x \implies x > \frac{c(\omega)}{2^{D(\omega)} - 3^m}.$$

This ensures that elements of the residue class indeed descend under the accelerated map.

5 Increment Grammar and Horizontal-Sum Construction

Remark (Increment Grammar from Minimal Denominator Growth)**.** The first difference sequence of A020914, given by A022921, induces a binary *increment grammar* $\{s, d\}$ that describes the growth of the *minimal admissible* total division count as the number of odd steps increases.

Each symbol records whether the minimal total division count increases by one (s) or two (d) when extending an admissible word by one additional odd step. Importantly, this grammar *does not* encode the realized parity word entries d_i (which may be arbitrarily large), but rather governs the evolution of the minimal denominator required for admissibility.

The increment grammar underlies the *horizontal-sum construction* used to enumerate all convergent parity words, admitting a canonical tabular realization. Specifically, admissible local patterns within the $\{s, d\}$ grammar determine the combinatorial growth of $\widetilde{A}(m)$ independently of the particular values of the parity word entries.

Proposition 5.1. The number of convergent parity words of length m is given by A186009.

Remark (Canonical Tabular Realization)**.** A canonical realization of this counting sequence via the horizontal-sum construction, together with its derivation from the increment grammar induced by A022921, is given in Appendix D. The appendix serves as a structural reference: the arguments in this section rely only on the grammar itself and not on any numerical properties of the tabulation.

Lemma 7 (Single-to-Double Row Sum Bound)**.** Consider a row produced by a *single* increment step with strictly positive entries and total sum S (i.e., the sum of its entries in the horizontal-sum table). If a *double* increment is applied immediately afterward, producing a row whose terminal entry equals the sum of the previous row and all other entries remain strictly positive, then the total sum of the resulting row is strictly less than $3S$.

Proof. By definition, the total sum of the single row equals S. Under a subsequent double step, the terminal entry of the new row is formed as the sum of all entries of the preceding row, and the previous terminal entry is duplicated.

Since the terminal entry of every row is the sum of the preceding row and is therefore strictly positive, the interior entries constitute a proper subset of the row's mass, and their total is strictly less than S.

Thus the double row consists of:

1. The interior entries of the single row, whose total is $S_{\text{int}} < S$;
2. One terminal entry equal to S; and
3. A duplicated terminal entry, also equal to S.

Hence the total sum of the double row is bounded by

$$S_{\text{int}} + S + S < 3S.$$

$\square$

Remark (Controlled Maximal Growth)**.** This lemma ensures that doubling steps contribute at most a controlled multiplicative factor to horizontal-sum growth, independent of the global cycle structure. Consequently, all admissible local patterns within a finite cycle admit finite maximal growth factors, which can be explicitly enumerated (see Appendix C). These bounds are key to establishing the vanishing density of residue classes associated with increasingly long stopping times.

6 Finite-State Linear Structure of the Horizontal-Sum Dynamics

This section formalizes the horizontal-sum construction as a finite-state positive linear dynamical system. The purpose is to show that the dynamics is order-preserving on a positive cone, so that rows realizing extremal local growth at any given stage continue to dominate all other admissible rows under every future evolution. Consequently, the extremal local growth calculations of Appendix C provide *uniform bounds* valid for all admissible rows, not merely for illustrative examples.

6.1 Linear Operator Formulation

Let $v_n = (a_1(n), a_2(n), \ldots, a_{k_n}(n))$ denote the generating row at level n, where the length k_n depends on the increment history. Let

$$S_n = \sum_i a_i(n)$$

be its total sum. Ignoring terminal duplication for the moment, the recurrence

$$a_i(n) = a_i(n-1) + a_{i-1}(n)$$

(with the convention that $a_0(n) = 0$) is linear with nonnegative coefficients.

A single increment step therefore corresponds to multiplication by a linear operator T_s with nonnegative entries. A double step duplicates the terminal entry after this linear evolution. This operation is also linear and can be represented by another operator T_d whose matrix entries are again nonnegative.

A single increment acts as a positive linear endomorphism $T_s^{(k)} : \mathbb{R}^k \to \mathbb{R}^k$, preserving row length, whereas a double increment acts as a positive linear map $T_d^{(k)} : \mathbb{R}^k \to \mathbb{R}^{k+1}$, increasing the row length by one via terminal duplication.

Hence each increment symbol $\sigma_n \in \{s, d\}$ induces a positive linear map

$$v_{n+1} = T_{\sigma_n} v_n,$$

and any admissible increment word corresponds to a product of matrices

$$v_{n+k} = T_{\sigma_{n+k-1}} \cdots T_{\sigma_n} v_n,$$

all of whose entries are nonnegative.

6.2 Monotone Extremal Propagation

Lemma 8 (Monotone Extremal Propagation)**.** Let v, w be two admissible rows of equal length satisfying $v_i \le w_i$ for all i. Then for any admissible increment sequence,

$$T_{\sigma_{n+k-1}} \cdots T_{\sigma_n} v \ \le \ T_{\sigma_{n+k-1}} \cdots T_{\sigma_n} w \quad \text{coordinatewise},$$

and consequently

$$\sum_i (Tv)_i \le \sum_i (Tw)_i.$$

Proof. Each operator T_s and T_d has only nonnegative matrix entries. Thus if $v \le w$ coordinatewise, then $T_s v \le T_s w$ and $T_d v \le T_d w$. Induction over the length of the increment word gives the result. □

This lemma shows that the horizontal-sum dynamics is *order-preserving*. Rows that maximize growth at one stage continue to dominate all others under every admissible future evolution.

6.3 Reduction to Local Pattern Bounds

Let $P = (\sigma_1, \ldots, \sigma_k)$ be an admissible local pattern of length k, with associated operator product

$$T_P = T_{\sigma_k} \cdots T_{\sigma_1}.$$

Define

$$\alpha(P) = \sup_{v \neq 0} \frac{\sum_{i=1}^{k_P} (T_P v)_i}{\sum_{i=1}^{k_0} v_i},$$

where k_0 and k_P are the row lengths before and after applying T_P, respectively.

Remark (Why extremal rows suffice)**.** The *positive cone* in $\mathbb{R}^m$ is

$$\mathbb{R}^m_{\ge 0} = \{v \in \mathbb{R}^m : v_i \ge 0\}.$$

The operators T_s and T_d are linear, homogeneous, and have only nonnegative entries. Hence for any pattern P, the operator T_P maps the positive cone into itself and is monotone with respect to the coordinatewise order.

Remark (Reduction to the simplex)**.** Because both the numerator and denominator in the definition of $\alpha(P)$ are positive linear functionals, the ratio

$$\frac{\sum_i (T_P v)_i}{\sum_i v_i}$$

is invariant under positive scaling of v. Hence the supremum may be taken over the normalized simplex

$$\Delta = \left\{ v \in \mathbb{R}^{k_0}_{\geq 0} : \sum_i v_i = 1 \right\},$$

where k_0 is the length of the input row. On Δ, the ratio reduces to a linear functional:

$$\sum_i (T_P v)_i = \sum_j \left(\sum_i (T_P)_{ij} \right) v_j.$$

Since Δ is a compact convex polytope, the maximum of this linear functional is attained at an extreme point of Δ, which corresponds to a coordinate vector concentrating all mass in a single coordinate j.

Thus the supremum defining $\alpha(P)$ is realized by rows supported on coordinates j for which the column sum $\sum_i (T_P)_{ij}$ is maximal. These rows are exactly those that saturate the growth inequalities of Lemma 7.

By positivity and Lemma 8, the supremum is attained by rows that saturate the inequalities in Lemma 7. Hence the extremal examples constructed in Appendix C provide valid upper bounds for $\alpha(P)$ for every admissible occurrence of P.

6.4 Maximal Growth via Grammar Cycles

Any admissible infinite increment sequence is a concatenation of grammar-allowed finite blocks. By definition of $\alpha(P)$, for any row v,

$$\sum_i (T_P v)_i \leq \alpha(P) \sum_i v_i.$$

Applying this inequality successively to the blocks in a concatenated increment word (whose total length is k) gives

$$\frac{S_{n+k}}{S_n} \leq \prod_{P \subset \text{word}} \alpha(P).$$

Therefore the asymptotic per-step growth rate satisfies

$$\limsup_{n\to\infty} S_n^{1/n} \leq \max_{\text{admissible cycles } C} \left(\prod_{P \in C} \alpha(P) \right)^{1/|C|}.$$

This reduces global growth control to the finite enumeration of local patterns performed in Appendix C, leading to the bound $\rho_5 < 3$ used in Theorem 11.

7 Cumulative Density of Convergent Classes

Remark (Roadmap)**.** In this section we combine the constructions developed in Sections 4 and 5 to analyze the asymptotic density of convergent residue classes. First, we record the number of admissible residue classes of stopping time i via the sequence $\widetilde{A}(i) = A186009(i+1)$ and the associated minimal denominator growth $B(i) = A020914(i)$. Then, using uniform bounds on $2^{B(i)}$ and the grammar-enforced suppression of locally expansive horizontal-sum growth, we show that the per-step growth of $\widetilde{A}(i)$ is strictly dominated by the denominator. Finally, this establishes that the density contribution of newly appearing convergent residue classes vanishes as $i \to \infty$, implying that almost every positive integer belongs to at least one convergent residue class.

7.1 Density Contributions and Denominator Growth

Define

$$\widetilde{A}(i) := A186009(i+1),$$

so that $\widetilde{A}(i)$ counts convergent residue classes of stopping time i.

Let

$$B(i) := A020914(i),$$

so that the density contribution of convergent residue classes of stopping time i is

$$\widetilde{A}(i)\, 2^{-B(i)}.$$

By definition of A020914 as the minimal exponent with $2^{B(i)} > 3^i$, the function $B(i)$ satisfies the following properties:

- $B(i)$ is strictly increasing, with increments either 1 or 2 (never zero).
- By construction, $2^{B(i)} > 3^i$ for all $i \geq 0$.
- Consecutive powers of two differ by a factor of 2, so

$$3^i < 2^{B(i)} < 2 \cdot 3^i.$$

Because $B(i)$ is defined as the minimal integer with $2^{B(i)} > 3^i$, the quantity $2^{B(i)}$ is the smallest power of two exceeding 3^i.

Remark (Indexing and Effective Growth Delay)**.** The OEIS sequences A020914 and A186009 employ differing indexing conventions. In order to compare numerator and denominator growth on a common scale, we define $\widetilde{A}(i) := A186009(i+1)$, so that the index i corresponds uniformly to stopping time (equivalently, parity-word length).

While the first doubling increment in A020914 occurs at index 1, its effect on $\widetilde{A}$ appears after a fixed finite delay of two growth steps. This delay is independent of i and does not affect the asymptotic growth comparison between numerator and denominator, which is controlled by the increment grammar introduced in Section 5.

Constant-factor control. This bound depends only on the definition of A020914 and does not rely on any delayed-doubling or numerator-growth arguments. Consequently, the reciprocal satisfies

$$2^{-B(i)} \in \left(\frac{1}{2\cdot 3^i}, \frac{1}{3^i}\right),$$

providing a *uniform constant-factor bound* for the density contribution at stopping time i.

We now record this observation formally, as it provides the fixed exponential scale against which numerator growth will be compared.

Lemma 9 (Uniform Control of Density Contributions)**.** For all $i \geq 0$,

$$3^i < 2^{A020914(i)} < 2 \cdot 3^i.$$

Consequently, the density contribution of convergent residue classes of stopping time i, given by $\widetilde{A}(i)2^{-A020914(i)}$, satisfies

$$\frac{\widetilde{A}(i)}{2\cdot 3^i} \;<\; \widetilde{A}(i)\,2^{-A020914(i)} \;<\; \frac{\widetilde{A}(i)}{3^i}.$$

Proof. By definition, $A020914(i)$ is the minimal integer m such that $2^m > 3^i$. Consecutive powers of two differ by a factor of 2, so

$$3^i < 2^{A020914(i)} < 2 \cdot 3^i.$$

Taking reciprocals and multiplying by $\widetilde{A}(i)$ gives the stated bounds. □

Lemma 10 (Disjointness of Minimal Convergent Classes)**.** For each $i \geq 0$, the residue classes counted by

$$\widetilde{A}(i) = A186009(i+1)$$

correspond to congruence classes of *minimal stopping time* i and are pairwise disjoint. Moreover, no positive integer belongs to two distinct minimal stopping-time classes.

Proof. Each admissible parity word of length i determines a residue class modulo $2^{B(i)}$ consisting of integers whose first descent below their initial value occurs precisely at step i. Minimality of the stopping time excludes the possibility that such an integer satisfies a shorter admissible parity word. Distinct admissible parity words of length i yield distinct residue classes modulo $2^{B(i)}$, and therefore their associated sets of integers are disjoint. Since each class is realized modulo the same modulus $2^{B(i)}$, disjointness as congruence classes implies disjointness as subsets of $\mathbb{N}$. $\square$

7.2 Growth Rate of the Number of Convergent Classes

Remark (Grammar-Enforced Suppression of Expansive Growth)**.** The proof of Theorem 11 does not rely on the absence of locally expansive configurations. Certain increment patterns are expansive in isolation, but the increment grammar forbids them or requires surrounding contractive structure.

(1) Local expansivity versus admissibility. A single double increment $\{d\}$ or the short pattern $\{s, d, d\}$ produces horizontal-sum growth with geometric mean exceeding 1. However, neither pattern is admissible under the increment grammar induced by A022921, as all admissible sequences must begin with $\{s\}$ and no occurrence of $\{d, d\}$ may appear without sufficient separating structure.

(2) Minimal realization of expansion. The shortest admissible pattern containing a double increment dd is the length-5 block

$$\{s, d, s, d, d\},$$

which constitutes the minimal expansive configuration. All admissible patterns of length ≤ 4 are contractive.

(3) Forced contractive buffering. The grammar enforces that the expansive 5-cycle cannot appear in isolation. The minimal admissible prefix leading to a $\{d, d\}$ occurrence is the length-7 block

$$\{s, d, s, d, s, d, d\},$$

which is contractive. Concatenating the expansive 5-cycle with this prefix yields a length-12 block that is also contractive. Thus expansive behavior is always dominated by mandatory contractive buffers.

(4) Consequence for asymptotic growth. Any admissible infinite increment sequence decomposes into finite grammar-allowed blocks. Horizontal-sum growth is submultiplicative under concatenation, so the asymptotic per-step growth rate is bounded above by the largest geometric mean among admissible blocks. Expansive blocks are separated by uniformly bounded contractive buffers, ensuring their asymptotic density in any admissible sequence is strictly less than one.

Theorem 11 (Maximal Admissible Growth Rate)**.** Let $\widetilde{A}(i) = A186009(i+1)$ denote the number of convergent residue classes of stopping time i, generated by the horizontal-sum construction associated with the increment grammar of A022921. Let $B(i) := A020914(i)$ denote the cumulative exponent of 2 up to level i.

The per-step growth of $\widetilde{A}(i)$ is bounded by the maximal geometric growth rate of a 5–cycle (see Appendix C, Equation (C.2)):

$$\limsup_{i\to\infty} \widetilde{A}(i)^{1/i} \leq \rho_5 < 3,$$

and consequently

$$\frac{\widetilde{A}(i)}{2^{B(i)}} \longrightarrow 0 \quad \text{as } i \to \infty.$$

Proof. The increment sequence A022921 induces a finite symbolic grammar on $\{s, d\}$, corresponding to single and double increments in the horizontal-sum construction.

Among all admissible sequences, the shortest potentially expansive configuration is the *5-cycle*

$$\{s, d, s, d, d\}.$$

All admissible words of length ≤ 4 produce horizontal-sum growth factors with geometric mean ≤ 1. A complete enumeration of admissible local patterns and their growth factors is given in Appendix C.

Explicit computation of the net horizontal-sum growth factor for the 5-cycle gives

$$\rho_5 := \big(\text{horizontal sum multiplier over the 5-cycle}\big)^{1/5} < 3.$$

Any admissible infinite increment sequence decomposes into concatenations of finite blocks permitted by the grammar. As explained in Remark 7.2, all locally expansive blocks are separated by mandatory contractive buffering. Horizontal-sum growth is submultiplicative under concatenation, so the asymptotic per-step geometric growth is bounded above by the maximal geometric mean among admissible blocks, which is attained by the 5-cycle and denoted ρ_5. Hence

$$\limsup_{i\to\infty} \widetilde{A}(i)^{1/i} \le \rho_5 < 3.$$

From Lemma 9, the denominator satisfies

$$3^i < 2^{B(i)} < 2 \cdot 3^i.$$

Consequently, there exists a constant $C > 0$, depending only on initial conditions and finite prefixes, such that

$$\widetilde{A}(i) \le C\,\rho_5^i \quad \text{for all } i.$$

Dividing by $2^{B(i)}$ gives

$$\frac{\widetilde{A}(i)}{2^{B(i)}} \le \frac{C\,\rho_5^i}{3^i} \longrightarrow 0,$$

which completes the proof. □

7.3 Vanishing of the Cumulative Density

The preceding results show that although the number of convergent residue classes $\widetilde{A}(i)$ grows with stopping time, its growth rate is strictly dominated by the denominator $2^{B(i)}$, which grows asymptotically like 3^i.

By Lemma 9, the density contribution at level i satisfies

$$\frac{\widetilde{A}(i)}{2 \cdot 3^i} < \widetilde{A}(i)\,2^{-B(i)} < \frac{\widetilde{A}(i)}{3^i}.$$

By Theorem 11, we have

$$\frac{\widetilde{A}(i)}{3^i} \longrightarrow 0 \quad \text{as } i \to \infty.$$

Although the per-level density contribution already tends to zero, the stronger geometric bound proved in Theorem 11 implies summability of the contributions, which is the key fact for the cumulative density argument below.

Lemma 12 (Summability of Density Contributions)**.** The series

$$\sum_{i=0}^{\infty} \widetilde{A}(i)\,2^{-B(i)}$$

converges.

Proof. By Theorem 11, there exists $C > 0$ such that

$$\widetilde{A}(i) \leq C\, \rho_5^i,$$

with $\rho_5 < 3$. By Lemma 9,

$$2^{B(i)} > 3^i.$$

Hence

$$\widetilde{A}(i)\, 2^{-B(i)} \leq C \left(\frac{\rho_5}{3}\right)^i.$$

Since $\rho_5/3 < 1$, the right-hand side is a geometric sequence with ratio less than 1, and therefore the series converges. □

Because the density contributions form a convergent series (Lemma 12), the tail of the series can be made arbitrarily small. Hence for any $\epsilon > 0$, there exists i_0 such that

$$\sum_{i \geq i_0} \widetilde{A}(i)\, 2^{-B(i)} < \epsilon.$$

In other words, the set of positive integers that fail to belong to any residue class of minimal stopping time has asymptotic density zero. By the descent properties established in Section 4, membership in such a class implies eventual decrease under the map.

Equivalently, almost every positive integer lies in at least one convergent residue class with respect to natural density, showing that convergent behavior dominates the Collatz map in a measure-theoretic sense.

Remark on growth bounds. The preceding argument is purely measure-theoretic and does not rely on lower bounds for the growth of $\widetilde{A}(i)$. Nevertheless, it is possible to establish a nontrivial exponential lower bound on the growth rate of $\widetilde{A}(i)$ by comparison with the auxiliary sequence A047749, which admits a closed-form binomial representation. This comparison shows that the exponential growth rate of $\widetilde{A}(i)$ is bounded below by $\sqrt{27/4}$. The derivation of this bound is independent of the density argument and is developed separately in Appendix F.

8 Global Descent in Density

8.1 Descent Words and Partition of the Descending Set

Remark (Minimal Descent Indexing)**.** Minimal descent is defined with respect to m, i.e., the first time an accelerated iterate satisfies $f^m(x) < x$. All residue classes R_ω in this section are indexed by m, so summations over $\mathcal{W}_{\min}$ correctly account for density.

Definition 8.1 (Descent Word)**.** An admissible parity word ω of length m is called a *descent word* if for all integers x in its associated residue class R_ω,

$$F_\omega(x) < x.$$

Definition 8.2 (Minimal Descent Word)**.** A descent word ω is *minimal* if no proper prefix of ω is itself a descent word.

Lemma 13 (Prefix Descent Sufficiency)**.** If ω is a descent word and η is any extension of ω (i.e. $\eta = \omega \star \tau$ for some nonempty suffix τ), then the residue class R_η is a subset of R_ω. Consequently, all integers realizing η have already descended at the prefix stage ω.

Here, $|\omega| = m$ counts the number of odd steps in the accelerated map f.

Proof. Any integer x realizing η realizes ω during its first $|\omega| = m$ accelerated iterations. Thus $x \in R_\omega$, so $R_\eta \subseteq R_\omega$. Since ω is a descent word, $F_\omega(x) < x$, and descent occurs before the suffix τ is applied. □

Theorem 14 (Minimal Descent Partition)**.** Let $\mathcal{W}_{\min}$ denote the set of minimal descent words. Then:

1. **Uniqueness.** Every integer x that eventually descends (i.e., there exists a finite m such that $f^m(x) < x$ under the accelerated map) admits a unique minimal descent word $\omega \in \mathcal{W}_{\min}$ such that $x \in R_\omega$.
2. **Disjointness.** If $\omega_1, \omega_2 \in \mathcal{W}_{\min}$ with $\omega_1 \neq \omega_2$, then
$$R_{\omega_1} \cap R_{\omega_2} = \varnothing.$$
3. **Exhaustion.** The set of all integers that ever descend equals
$$\bigcup_{\omega \in \mathcal{W}_{\min}} R_\omega.$$

Proof. (1) For any x that descends, consider the first time m such that $f^m(x) < x$. The parity word of length m realized by x is a descent word, and no shorter prefix can be, so it is minimal.

(2) If $x \in R_{\omega_1} \cap R_{\omega_2}$, then x realizes both parity words, so their realized parity sequences agree up to the minimum of their lengths. Minimality forces $\omega_1 = \omega_2$.

(3) Every descending integer has a minimal descent prefix by (1), and Lemma 13 ensures extensions produce no new elements. □

Corollary 15 (No Overcounting of Density)**.** The residue classes R_ω for $\omega \in \mathcal{W}_{\min}$ are pairwise disjoint. Hence

$$\mu\left(\bigcup_{\omega \in \mathcal{W}_{\min}} R_\omega\right) = \sum_{\omega \in \mathcal{W}_{\min}} \mu(R_\omega),$$

and extensions of descent words contribute no additional density.

All minimal descent words are indexed by m, and any extension $\eta = \omega \star \tau$ does not add new integers; thus summing over $\mathcal{W}_{\min}$ correctly accounts for the total density.

8.2 Asymptotic Global Descent of Almost Every Integer

Theorem 16 (Asymptotic Global Descent)**.** The union of convergent residue classes has asymptotic density one. Consequently, almost every positive integer eventually reaches a smaller value under iteration of the accelerated Collatz map, in the sense of natural density.

Proof. Let $\widetilde{A}(i) = A186009(i+1)$ denote the number of convergent residue classes of stopping time i (accelerated map), and let $B(i) = A020914(i)$ denote the cumulative exponent of 2 at level i. By Theorem 11, we have

$$\frac{\widetilde{A}(i)}{2^{B(i)}} \longrightarrow 0 \quad \text{as } i \to \infty.$$

By construction, each residue class consists of integers that descend after finitely many accelerated iterations. Lemma 12 shows that the density contributions of convergent residue classes are summable, and hence that the cumulative density of all such classes converges. Since these classes partition the integers up to a null set, this limit must equal one. Therefore the set of integers not contained in any convergent residue class has asymptotic density zero.

Therefore, for almost every positive integer x (in the sense of natural density), there exists a finite m such that

$$f^m(x) < x,$$

where f denotes the accelerated Collatz map. $\square$

Remark (Exceptional Orbits)**.** This theorem does not preclude the existence of exceptional trajectories outside the convergent residue classes. Any such set must have asymptotic density zero, so the result establishes descent in a measure-theoretic sense rather than an absolute guarantee.

9 Hypothetical Zero-Density Trajectories

The density result established above does not exclude the logical possibility of non-convergent Collatz trajectories. However, it severely restricts the form that any such trajectory could take. In particular, any exceptional set must lie entirely outside the union of convergent residue classes whose cumulative asymptotic density approaches one, confining any hypothetical exception to a set of asymptotic density zero.

Consequently, an exceptional trajectory could not belong to any of the residue-class families constructed in the preceding sections, nor arise from the structured parity patterns that generate those classes. Any trajectory avoiding all convergent residue classes would also avoid all patterns admissible under the increment grammar, highlighting the structural rigidity imposed by the combinatorial framework.

In this sense, any hypothetical exception would be *dynamically isolated*: it would not be supported by a scalable arithmetic structure or by a residue-class hierarchy of positive density, but would instead persist only as a sparse, non-propagating phenomenon within the integer space.

10 Conclusion

In this work, we have analyzed the Collatz map via a structural decomposition of the positive integers into non-overlapping residue classes determined by finite parity patterns, each associated with a minimal stopping time. By enumerating convergent parity words and controlling the maximal admissible growth rate relative to the corresponding powers of two, we rigorously demonstrated that the cumulative density of convergent classes approaches one.

The analysis is further sharpened by showing that descent behavior admits a canonical minimal description: integers that eventually descend are partitioned into disjoint residue classes indexed by minimal descent words, and longer admissible parity patterns introduce no additional density. This reduces the global problem to a non-redundant symbolic skeleton governing all density-carrying trajectories.

This density-based approach establishes that almost every positive integer eventually reaches a strictly smaller value under the accelerated Collatz map. While the argument does not preclude zero-density exceptional trajectories, it provides a strong measure-theoretic framework for understanding the typical dynamics of the map.

Viewed abstractly, this perspective reframes the Collatz problem as an analysis of integer lattices partitioned by arithmetic residue classes, highlighting how parity- and residue-based combinatorial structures govern discrete dynamical systems. It emphasizes the distinction between "almost all" and "all," yielding a credible, rigorous statement fully consistent with the density analysis.

A Canonical Determination of the Constant $c(\omega)$

Each parity word ω induces an affine map $F_\omega(x)$ of length m (see Section 3)

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}}, \qquad D(\omega) = d_0 + \cdots + d_{m-1},$$

where $c(\omega)$ is uniquely defined modulo $2^{D(\omega)}$. In this appendix, we describe a canonical choice and a simple method to compute it.

Canonical Choice

For a fixed ω, any integer x in the corresponding residue class satisfies

$$3^m x + c \equiv 0 \pmod{2^{D(\omega)}}.$$

We remove ambiguity by selecting the unique representative

$$0 \leq c(\omega) < 2^{D(\omega)}.$$

This choice depends only on ω and is independent of the representative x.

Step-by-Step Computation

Given ω of length m and total division count $D(\omega)$:

1. Choose any integer x in the residue class corresponding to ω.
2. Compute $y = 3^m x$.
3. Let $r = \lceil y/2^{D(\omega)} \rceil$ be the smallest integer such that $r \cdot 2^{D(\omega)} \geq y$.
4. Define $c(\omega) = r \cdot 2^{D(\omega)} - y$.

By this construction, $0 \leq c(\omega) < 2^{D(\omega)}$ ensuring that

$$F_\omega(x) = \frac{3^m x + c(\omega)}{2^{D(\omega)}} \in \mathbb{Z} \quad \text{for all } x \text{ in the class.}$$

Example

Let $\omega = (1, 1, 1, 4)$ with $m = 4$ and $D(\omega) = \sum_{i=0}^{3} d_i = 1 + 1 + 1 + 4 = 7$:

1. Choose $x = 15$. Compute $y = 3^4 \cdot 15 = 1215$.
2. $2^7 = 128$, so $r = \lceil 1215/128 \rceil = 10$.
3. $c(\omega) = 10 \cdot 128 - 1215 = 65$.

Hence the canonical map is

$$F_\omega(x) = \frac{81x + 65}{128}.$$

Checking another representative $x = 143$:

$$F_\omega(143) = \frac{81 \cdot 143 + 65}{128} = 91,$$

confirming that the same $c(\omega)$ works for the entire residue class.

Remark

This construction is purely illustrative and is not required for the density or extremal growth arguments in the main text. It demonstrates concretely how the affine map is fully determined by the parity word ω.

Since any two integers realizing the same parity word differ by a multiple of $2^{D(\omega)}$, the value of $c(\omega)$ computed by this procedure is independent of the chosen representative.

B Extremal Local Growth Analysis

Lemma 17 (Local Growth Bounds)**.** For each admissible local pattern occurring within A022921, the ratio of the total sum of entries after applying the pattern to the total sum before applying it is bounded above as follows, uniformly over all admissible placements of (delayed) doubling within the pattern:

Pattern	Max Growth Factor
s, d	3
d, d	123/33
s, d, s	341/131
d, d, s	329/123

Here s denotes a single increment and d a double increment in the horizontal-sum construction. Note that these bounds are independent of the starting row and only depend on the local pattern.

Proposition B.1 (Extremal 5-Cycle Growth)**.** The maximal geometric growth rate achievable by any admissible infinite word under the horizontal-sum dynamics associated with A186009 is bounded by

$$\rho_5 = \left(3^2 \cdot (123/33)^1 \cdot (341/131)^1 \cdot (329/123)^1\right)^{1/5} \approx 2.976 < 3,$$

where the exponents record the number of times each admissible local pattern appears within a single admissible 5-cycle (the minimal admissible expansive block). See Section C.1 for explicit enumeration: two occurrences of $\{s, d\}$, one of $\{d, d\}$, one of $\{s, d, s\}$, and one of $\{d, d, s\}$.

This enumeration ensures that every admissible infinite word can be decomposed into concatenations of 5–cycles whose aggregate growth is controlled by ρ_5. See Lemma 18 for a rigorous argument that 5-cycles cannot be avoided and therefore this bound applies globally.

Lemma 18 (Inevitable 5-Cycle Insertion)**.** Every contractive sequence of cycles under the admissibility grammar of A022921 necessarily contains at least one embedded 5-cycle within either a 12-cycle (7-cycle + 5-cycle) or as a standalone 5-cycle following sufficient consecutive 12-cycles.

In particular:

1. Purely consecutive 7-cycles cannot occur.
2. Expansive 5-cycles appear at bounded intervals within any admissible infinite word.
3. After sequences of three or four consecutive 12-cycles (yielding 41- or 53-cycle blocks), the grammar forces a 5-cycle insertion without exceeding the contractive bound of unity.
4. It is therefore impossible to indefinitely avoid 5-cycles; they are structurally unavoidable.

Remark (Extremal 5-Cycle Envelope)**.** Although the 12-cycle is contractive overall, it contains a 5-cycle in its tail. Repeated sequences of 12-cycles may occasionally yield consecutive 5-cycles once enough contractive slack has accumulated, as seen in the 41-cycle ($3 \times 12 + 5$) or 53-cycle ($4 \times 12 + 5$) structures, but the grammar ensures that 5-cycles cannot be skipped indefinitely.

This observation justifies treating an infinite repetition of 5-cycles as a strict upper envelope: it saturates all local growth factors, while any admissible word, including A186009, contains interspersed contractive 12-cycles, so its per-symbol growth is strictly below the 5-cycle maximal rate ρ_5.

Remark (Connection to Theorem 11)**.** The quantity ρ_5 defined above represents the maximal geometric growth factor per symbol for any admissible word under the horizontal-sum dynamics. Taking the fifth root over one full 5-cycle yields the maximal per-symbol growth, which directly controls $\limsup_{i\to\infty} \widetilde{A}(i)^{1/i}$ in Theorem 11. Because $\rho_5 < 3$, the growth of the number of convergent residue classes is asymptotically strictly slower than the corresponding powers of two, guaranteeing that the density of integers with large stopping times vanishes.

Remark (Derivation and Attribution of Local Growth Constants)**.** The horizontal-sum framework used to analyze $\widetilde{A}(i)$ is inspired by the Pascal-type summation approach introduced by Winkler [8], which provided the original insight that admissible Collatz parity words can be organized into structured combinatorial growth processes.

The present work adapts this perspective to the specific setting of A022921 and A186009, with one key modification: unlike Winkler's vertical-sum formulation, the horizontal orientation used here is essential for preserving row-wise combinatorial structure, enabling precise extremal growth calculations. This extension allows a complete analysis of local growth configurations and the derivation of the uniform 5-cycle bound.

The constants shown in Lemma 17 are obtained by examining the horizontal-sum dynamics governing $\widetilde{A}(i)$. Within each local configuration, elements arising from single steps evolve according to standard Pascal-type summation, yielding maximal growth proportional to powers of 2. Additional entries introduced by doubling steps are bounded by summing on top of this baseline growth.

For each admissible local pattern occurring within A022921, all possible placements of delayed doubling contributions were examined, and the largest achievable ratio of post-pattern to pre-pattern row sums was recorded. No assumptions beyond the combinatorial structure of the horizontal-sum construction are used.

A complete derivation of the horizontal-sum mechanism, together with detailed computations of all local growth constants and their admissible concatenations, is given in Appendix C. Consequently, while the conceptual framework traces back to Winkler's original insight, the bounds employed here are established entirely within the present paper. Note that no analytic or probabilistic assumptions are made; all bounds arise from finite combinatorial enumeration.

It follows that any admissible sequence under A022921, including those relevant for A186009, is strictly dominated in per-symbol growth by a concatenation of 5-cycles, yielding the uniform bound of Proposition B.1.

C Extremal Local Growth Calculations

This appendix provides the explicit finite computations underlying the local growth bounds used in the density argument. Recall that the sequence A020914 has first differences given by A022921, consisting of repeated blocks of *single* steps (s) and *double* steps (d).

C.1 Cycle Structure and Net Multiplicative Effect

A frequently occurring block is the 7–cycle

$$\{s, d, s, d, s, d, d\},$$

which contains 3 singles and 4 doubles. The corresponding number of factors of 2 is

$$3 + 2(4) = 11,$$

so the net multiplicative contribution of this cycle satisfies

$$\frac{2^{11}}{3^7} < 1.$$

Thus the 7–cycle is strictly contractive. Every 7–cycle is followed by a 5–cycle

$$\{s, d, s, d, d\},$$

which contains 2 singles and 3 doubles, contributing

$$2 + 2(3) = 8$$

factors of 2. Since

$$\frac{2^8}{3^5} > 1,$$

the 5–cycle is expansive.

Concatenating the 7–cycle and 5–cycle produces a 12–cycle

$$\{s, d, s, d, s, d, d, s, d, s, d, d\},$$

containing 5 singles and 7 doubles. The total number of factors of 2 is

$$5 + 2(7) = 19,$$

and hence

$$\frac{2^{19}}{3^{12}} < 1.$$

Therefore, the 12–cycle is again contractive.

Repeated 12-cycles remain contractive, suppressing growth until enough slack accumulates to allow a 5-cycle insertion. The 5-cycle represents the unique locally expansive pattern, though its repetition is constrained by the grammar.

To visualize the interaction between contractive and expansive cycles, Figure 1 depicts the state transitions of A022921. The red edges indicate the locally expansive 5-cycle, the blue edges the contractive 7-cycle, and purple edges highlight shared transitions.

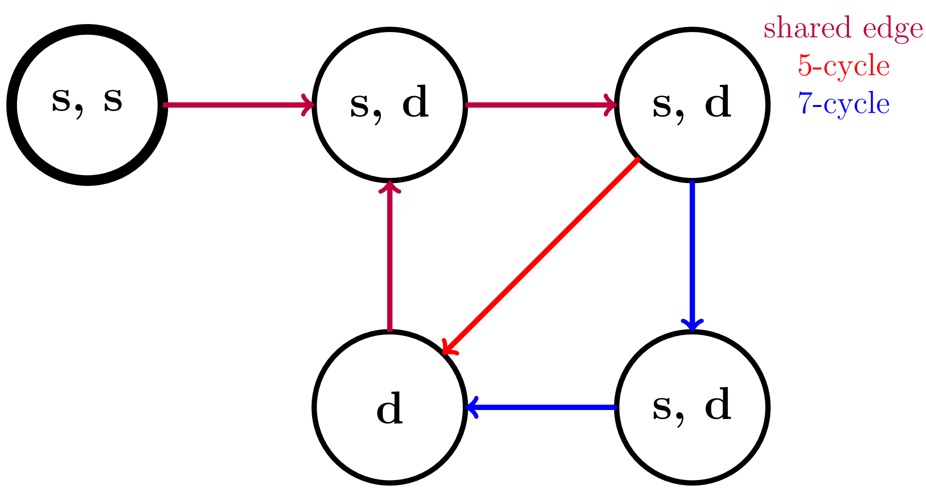

Figure 1: State Transition Diagram for A022921 highlighting cycles: contractive 7-cycle (blue), locally expansive 5-cycle (red), shared edges (purple). Lowercase letters reflect single (s) and double (d) parity steps consistent with the main text notation.

The figure highlights the locally expansive 5-cycle (red), the contractive 7-cycle (blue), and the shared edges (purple), visually illustrating the grammar constraints of the sequence A022921.

Horizontal-Sum Dynamics and Pascal-Type Growth

The generating rows evolve by a horizontal-sum rule:

$$a_i(n) = a_i(n-1) + a_{i-1}(n), \quad i, n \in \mathbb{N}, \tag{C.1}$$

so that sequences of single steps exhibit ordinary Pascal–type growth. Double steps differ in that the terminal entry is duplicated, causing the final term to appear twice while preserving monotonicity.

Because the grammar forces expansive configurations to appear only within this minimal 5-block, extremal asymptotic growth is realized by repeated alignment of this block, making it sufficient to analyze this configuration. Thus for extremal 5-cycle analysis, the local maximal sequence (dominating per-pattern growth) is

$$\{s, d, s, d, d\}, \{s, d, s, d, d\},$$

enumerating occurrences of each admissible local pattern within a 5-cycle.

Admissible Local Patterns

Within A022921, only four local configurations occur:

$$(s, d), \quad (d, d), \quad (s, d, s), \quad (d, d, s).$$

The associated coefficient counts within a 5-cycle are:

Pattern	Associated coefficient count
s, d	2
d, d	1
s, d, s	1
d, d, s	1

C.2 Illustrative Extremal Computations

The generating rows evolve by a horizontal–sum rule (see Equation C.1) so that sequences of single steps exhibit ordinary Pascal–type growth. Double steps differ only in that the terminal entry of the row is duplicated, causing the final term to appear twice in the generating tuple while preserving monotonicity of the preceding entries.

These examples are chosen so that each step saturates the inequalities of Lemma 7, thereby realizing extremal growth. Equalities shown below should be interpreted as extremal upper-bound realizations rather than identities valid for all rows.

The following row-by-row computations realize extremal growth for the admissible local patterns and together determine the sharpest uniform bounds used in Lemma 17. Each example illustrates how a local pattern can saturate its maximal growth factor under the horizontal-sum dynamics. Let S denote the total sum of entries in the generating row prior to a given local pattern.

Example 1.

$$\textit{Single } a(10) = \underbrace{173}_{\text{sum:}=S} : \underbrace{1, 7, 25, 55, 85}_{S}$$

$$\textit{Double } a(11) = \underbrace{476}_{\text{sum}=3S} : \underbrace{1, 8, 33, 88}_{S}\underbrace{173}_{S}, \underbrace{173}_{S}$$

$$\textit{Single } a(12) = \underbrace{961}_{\text{sum}=9S} : \underbrace{1, 9, 42, 130}_{2S}\underbrace{303}_{3S}, \underbrace{476}_{4S}$$

$$\textit{Double } a(13) = \underbrace{2652}_{\text{sum}=33S} : \underbrace{1, 10, 52, 182}_{4S}, \underbrace{485}_{7S}, \underbrace{961}_{11S}, \underbrace{961}_{11S}$$

$$\textit{Double } a(14) = \underbrace{8045}_{\text{sum}=123S} : \underbrace{1, 11, 63, 245}_{8S}, \underbrace{730}_{15S}, \underbrace{1691}_{26S}, \underbrace{2652}_{37S}, \underbrace{2652}_{37S}$$

$$\textit{Single } a(15) = \underbrace{17637}_{\text{sum}=329S} : \underbrace{1, 12, 75, 320}_{16S}, \underbrace{1050}_{31S}, \underbrace{2741}_{57S}, \underbrace{5393}_{94S}, \underbrace{8045}_{131S}$$

The resulting maximal local ratios are

Pattern	Max Ratio
s, d	3/1
d, d	123/33
s, d, s	9/3
d, d, s	329/123

Example 2.

$$\textit{Single } a(12) = \underbrace{961}_{\text{sum:=}S} : \underbrace{1, 9, 42, 130, 303, 476}_{S}$$

$$\textit{Double } a(13) = \underbrace{2652}_{\text{sum=}3S} : \underbrace{1, 10, 52, 182, 485}_{S}, \underbrace{961}_{S}, \underbrace{961}_{S}$$

$$\textit{Double } a(14) = \underbrace{8045}_{\text{sum=}13S} : \underbrace{1, 11, 63, 245, 730}_{2S}, \underbrace{1691}_{3S}, \underbrace{2652}_{4S}, \underbrace{2652}_{4S}$$

$$\textit{Single } a(15) = \underbrace{17637}_{\text{sum=}37S} : \underbrace{1, 12, 75, 320, 1050}_{4S}, \underbrace{2741}_{7S}, \underbrace{5393}_{11S}, \underbrace{8045}_{15S}$$

$$\textit{Double } a(16) = \underbrace{51033}_{\text{sum=}131S} : \underbrace{1, 13, 88, 408, 1458}_{8S}, \underbrace{4199}_{15S}, \underbrace{9592}_{26S}, \underbrace{17637}_{41S}, \underbrace{17637}_{41S}$$

$$\textit{Single } a(17) = \underbrace{108950}_{\text{sum=}341S} : \underbrace{1, 14, 102, 510, 1968}_{16S}, \underbrace{6167}_{31S}, \underbrace{15759}_{57S}, \underbrace{33396}_{98S}, \underbrace{51033}_{139S}$$

The corresponding maximal local ratios are

Pattern	Max Ratio
s, d	$3/1$
d, d	$13/3$
s, d, s	$341/131$
d, d, s	$37/13$

Improved Extremal Alignment

Because the total multiplicative growth over a cycle is the product of the local growth factors, the geometric mean over one full cycle is maximized by selecting the maximal admissible factor at each local pattern occurrence. Consequently, once a uniform upper bound has been established for each admissible local pattern, the extremal global growth rate over a complete cycle is obtained by multiplying these maximal local factors according to their frequencies and taking the corresponding geometric mean.

Equivalently, because total growth over any admissible word is multiplicative in its local factors, the asymptotic per-symbol growth rate is the geometric mean of these *factors; hence the maximal rate over the entire language generated by the grammar is achieved by the word that maximizes this mean subject to the grammar constraints.*

Each example yields a valid (not necessarily sharp) upper bound on the maximal growth factor associated with each admissible local pattern. Since these bounds arise from the same uniform recurrence and apply globally to all admissible configurations, the true maximal growth factor for a given pattern is bounded above by the smallest recorded bound among all such valid upper bounds.

Because the recurrence is monotone with respect to the placement of delayed doubling, no other admissible configuration can exceed these extremal realizations. Accordingly, combining the sharpest bounds obtained across different initiating rows yields the tightest global envelope on admissible local growth.

Pattern	Max Ratio	Associated coefficient count
s, d	3	2
d, d	123/33	1
s, d, s	341/131	1
d, d, s	329/123	1

Consequently, the maximal geometric growth rate of a 5–cycle satisfies

$$\Big(3^2 \cdot (123/33) \cdot (341/131) \cdot (329/123)\Big)^{1/5} \approx 2.976 < 3. \tag{C.2}$$

This bound directly feeds into Theorem 11 and ensures that the per-symbol growth of any admissible infinite sequence under A186009 is strictly dominated by 3, as explained in the main text.

Conclusion. Every admissible growth pattern of A186009 is constrained by the grammar of A022921. Infinite concatenations of 5-cycles, though forbidden, saturate local growth constraints and provide a strict upper envelope. Equation (C.2) confirms that even this extremal envelope has geometric growth rate < 3, ensuring subdominance relative to the denominator $2^{B(i)}$.

D Horizontal-Sum Table for A186009

For reference, a portion of A186009 generated by the horizontal-sum construction is given below as per an algorithm derived from Winkler [8]. This table is included for illustration of the horizontal-sum mechanism; none of the density or growth bounds in the main text depend on these specific numeric values.

Table 1: A186009 Horizontal Sum Formulation

n	$a_1(n)$	$a_2(n)$	$a_3(n)$	$a_4(n)$	$a_5(n)$	$a_6(n)$	$a_7(n)$	$a_8(n)$	$a_9(n)$	$\sum a_i(n)$
1	1									1
2	1									1
3	1									1
4	1	1								2
5	1	2								3
6	1	3	3							7
7	1	4	7							12
8	1	5	12	12						30
9	1	6	18	30	30					85
10	1	7	25	55	85					173
11	1	8	33	88	173	173				476
12	1	9	42	130	303	476				961
13	1	10	52	182	485	961	961			2652
14	1	11	63	245	730	1691	2652	2652		8045
15	1	12	75	320	1050	2741	5393	8045		17637
16	1	13	88	408	1458	4199	9592	17637	17637	51033
17	1	14	102	510	1968	6167	15759	33396	51033	108950

There is a sequence closely related to A186009 called A100982. A186009 prepends the number 1 to the A100982 sequence. Along with the listing of initial terms for A100982 comes an algorithm which explains how to generate the entire sequence starting with an initial value of 1. As a result of its close similarity, this method can also be leveraged to generate all of the terms in A186009.

Theorem 2 [8] referenced in A100982 asserts that the elements in the sequence can be derived in a Pascal–triangle–type manner, beginning with an initial condition:

$$a(1) = a_1(1) = 1$$

The algorithm for $A100982(n)$ states that the k^{th} term of the n^{th} series component can be computed as follows:

$$a_{k+1}(n) = a_k(n) + a_k(n-1); k \in \mathbb{N}_0, n \in \mathbb{N}$$

The algorithm also stipulates that the last term in the sequence is repeated whenever $A022921(n-2) = 2$ (which can be derived from A020914 by taking the first differences between its terms). In doing so, this imposes the implicit constraint that this rule applies only when $n \geq 2$. Keeping in mind that repeated terms in A022921 occur only once $n \geq 2$, the recurrence is extended to $n = 1$ by treating the duplication condition as inactive at the boundary, producing an augmented vertical-sum expansion.

In rows without terminal duplication, this recurrence is exactly Pascal's binomial rule; duplication acts only at the boundary and therefore preserves the monotone, binomial-type growth of interior entries. This boundary-only modification is precisely why the extremal growth analysis in Appendix C reduces to *local pattern frequencies* rather than global row shape.

We reformulate this construction such that the first non-zero term in each row is always at index 1 and thus the formula henceforth (producing the same expansion components up to a reindexing of coordinates) is modified as follows.

Preserving $a(1) = a_1(1) = a_1(n) = 1$ for all $n \in \mathbb{N}$, the i^{th} term of the n^{th} A100982 element is:

$$a_i(n) = a_i(n-1) + a_{i-1}(n); i, n \in \mathbb{N}$$

This reindexing does not change any row sums or terminal-duplication positions; it only relocates coordinates. Using this formulation the initial term for all values of $n \in \mathbb{N}$ is always 1 rather than shifting out for each value of n. The other significant difference is that in this formulation the values for $a_i(n)$ appear horizontally as opposed to vertically. The doubling rule for the terminal entry remains unchanged in substance – namely that it occurs when $A022921(n-2) = 2$ - which is precisely the single/double grammar governing admissible growth in the preceding appendices, and hence the local multiplicative structure used in the extremal bounds.

Because A186009 is obtained from A100982 by prepending an initial row, all row indices shift by one. Consequently the terminal-duplication condition expressed in the original formulation as $A022921(n-2) = 2$ now becomes $A022921(n-3) = 2$ in the present indexing.

E Enumeration of Admissible Parity Words

Purpose and Scope

This appendix provides an explicit, constructive procedure for enumerating admissible parity words of fixed length under the prefix constraints governed by A020914. While none of the density or growth bounds in the main text depend on these constructions, the algorithm clarifies the combinatorial structure underlying A186009 and makes explicit the finite grammar inducing admissible parity words implicit in the horizontal-sum formulation.

Prefix-Sum Formulation

Let a parity word of length $m \geq 1$ be a tuple

$$(d_0, d_1, \ldots, d_{m-1}), \qquad d_i \in \mathbb{N},$$

where d_i records the number of divisions by 2 following the ith odd step. Define the prefix sums

$$H(i) = \sum_{j=0}^{i} d_j.$$

Admissible parity words of length m are exactly those satisfying

$$H(i) < A020914(i+1) \quad (0 \leq i < m-1), \qquad H(m-1) = A020914(m).$$

The strict inequalities for $i < m-1$ suppress shorter stopping-time words, while equality in the final column ensures the total division count matches the minimal exponent associated with length m.

Thus admissible parity words are determined by a bounded prefix-sum search with backtracking.

Constructive Enumeration Procedure

Fix a word length $m \geq 1$ and define m columns numbered $i = 0, \ldots, m-1$. Below each i calculate the parity word grammar bounds $A(i+1) = A020914(i+1)$.

Then, under each column, place the exponent entries d_i of the parity word. The left-to-right horizontal sum $H(i)$ is the sum of entries up to column i, where $H(i)$ must satisfy the admissibility bounds described above.

Iterative Search

Generation of the set of all parity words of length m proceeds by iterating through the following steps until all admissible words have been found.

Set $i = 0$ and $H(-1) = 0$ as the initial boundary condition. Begin at step 1.

1. Set successive row entries $d_i = 1$ up to column $m - 2$:
   (a) If $i < m - 1$, set $d_i = 1$. Increment i and repeat Step 1.
   (b) Otherwise, proceed to Step 2.
2. Calculate the final entry d_{m-1} so that the horizontal sum matches $A(m)$:
$$d_{m-1} = A(m) - H(m - 2),$$
   record the resulting parity word $(d_0, \dots, d_{m-1})$. Continue to step 3.
3. Increment the $(i - 1)^{th}$ column if this is admissible by the grammar:
   (a) If $H(i - 1) < A(i) - 1$, copy the previous row entries up to $i - 2$.
   Increment d_{i-1} by 1, and return to Step 1.
   (b) Otherwise, proceed to Step 4.
4. Check the termination condition for the parity word search:
   (a) If i is greater than 1, decrement i by one and return to Step 3.
   (b) Otherwise, all admissible parity words of length m have been generated. Stop.

Example: Parity Words of Length $m = 5$

For $m = 5$, the bounds are

$$A020914(1{:}5) = (2, 4, 5, 7, 8).$$

All admissible parity words must satisfy

$$H(0) < 2,\ H(1) < 4,\ H(2) < 5,\ H(3) < 7,\ H(4) = 8.$$

The search procedure yields the seven admissible words shown in Table 2.

Table 2: Admissible parity words of length $m = 5$.

i	0	1	2	3	4
$A(i+1)$	2	4	5	7	8
Word	d_0	d_1	d_2	d_3	d_4
1	1	1	1	1	4
2	1	1	1	2	3
3	1	1	1	3	2
4	1	1	2	1	3
5	1	1	2	2	2
6	1	2	1	1	3
7	1	2	1	2	2

Thus $A186009(m+1) = 7$ counts admissible parity words of length $m = 5$. The final word $(1, 2, 1, 2, 2)$ is extremal in the sense that its prefix sums approach the bounds $A020914(i + 1)$ as closely as possible for $i < m - 1$ while remaining admissible, and achieve equality at $i = m - 1$.

F A Binomial Lower Comparison for A186009

The sequence A047749 admits a horizontal-sum construction analogous to that of A186009, but with a stricter terminal rule: its doubling mechanism is governed by A000034($n-2$) rather than A022921. As a result, A047749 forbids *consecutive* doublings, while such events occur periodically in A186009. This structural restriction yields a natural comparison sequence.

Pointwise comparison

Because A047749 evolves under the same Pascal-type recursion but with a subset of the admissible doubling events, its row sums form a pointwise lower bound (up to index shift):

$$A047749(n) \leq A186009(n+1), \qquad n \geq 0.$$

The two sequences agree initially, but diverge once the first consecutive doubling occurs in A186009; thereafter the inequality is strict.

Closed form for A047749

Unlike A186009, the restricted doubling grammar of A047749 permits an exact description. Its terms split naturally into even and odd indices:

Even indices ($n = 2m$).

$$a(n) = \frac{1}{2m+1}\binom{3m}{m}.$$

Odd indices ($n = 2m+1$).

$$a(n) = \frac{1}{2m+1}\binom{3m+1}{m+1}.$$

Equivalently, using Gamma functions,

$$a(2m) = \frac{\Gamma(3m+1)}{\Gamma(2m+2)\Gamma(m+1)}, \qquad a(2m+1) = \frac{\Gamma(3m+2)}{\Gamma(2m+2)\Gamma(m+2)}.$$

Asymptotic growth

Let $\psi(x) = \Gamma'(x)/\Gamma(x)$ be the digamma function. Differentiating the logarithm of the Gamma representations gives

$$\frac{d}{dm} \ln a(2m) = 3\psi(3m+1) - 2\psi(2m+2) - \psi(m+1),$$
$$\frac{d}{dm} \ln a(2m+1) = 3\psi(3m+2) - 2\psi(2m+2) - \psi(m+2).$$

Using $\psi(x) = \ln x + O(1/x)$ as $x \to \infty$, both expressions have the same limit:

$$\lim_{m\to\infty} \frac{d}{dm} \ln a(n) = \ln\left(\frac{3^3}{2^2}\right) = \ln\left(\frac{27}{4}\right).$$

Thus

$$\lim_{n\to\infty} \frac{a(n+2)}{a(n)} = \frac{27}{4}, \qquad a(n) \sim C\left(\sqrt{\frac{27}{4}}\right)^n.$$

Implication for A186009

Since A047749 is generated by a subset of the admissible doubling events for A186009, it provides a constructive comparison sequence with explicit exponential growth rate

$$\sqrt{\frac{27}{4}} = \frac{3\sqrt{3}}{2} \approx 2.598.$$

Hence A186009 admits an explicit exponentially growing comparison sequence, and grows strictly faster once consecutive doubling regimes become active.

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